

ANOVA 1

Aliaksandr Hubin

Feb 24 2025

Topics for this lecture

- Introductory example
- “ANOVA”
- The F-test
- Estimation and parametrization
- Post hoc contrasts and pairwise tests

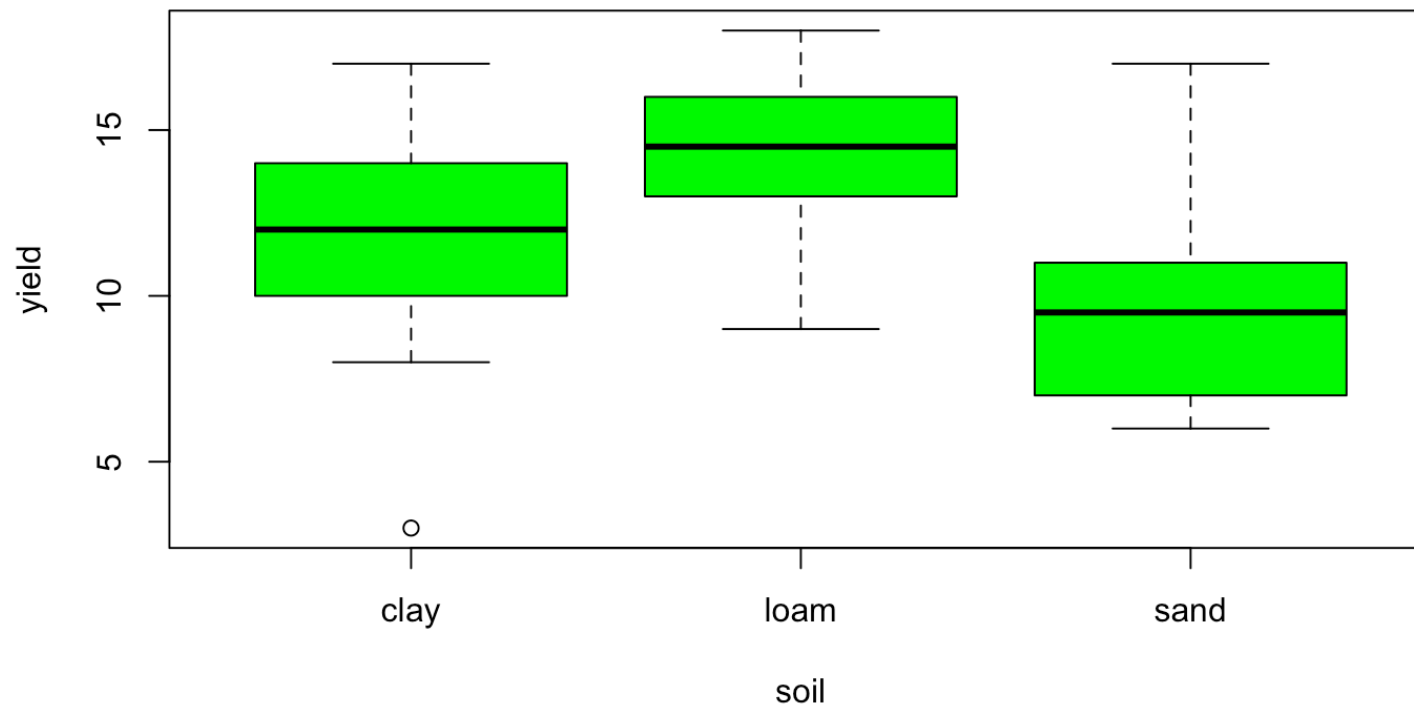
Objectives of lecture

- *Basic framework*: The response variable is continuous as for regression but the predictor variables are *factors* like fertilizer, soil type, country, ...
- *Basic idea*: The statistical analysis is based on studying the variability in data. Hence the name ANOVA:
 - ANalysis Of VAriance.
- Objectives as for regression: predict and explain

Example Yields

Yield from a crop experiment on $k = 3$ different soil types. There are $n_i = 10$ random replicates per soil type. A total of $n = k \cdot n_i = 30$ observations altogether.

Are the expected yields different?



Linear Regression Formulation

Model with dummy variables for soil type:

$$y_j = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \beta_3 x_{3j} + \epsilon_j, \epsilon_j \sim N(0, \sigma^2)$$

where:

- $x_{1j} = 1$ if soil is clay, else 0
- $x_{2j} = 1$ if soil is loam, else 0
- $x_{3j} = 1$ if soil is sand, else 0
- $\beta_0 = \mu$ = intercept (grand mean)
- $\beta_0 + \beta_1 = \mu_1$ = mean of yield for clay
- $\beta_0 + \beta_2 = \mu_2$ = mean of yield for loam
- $\beta_0 + \beta_3 = \mu_3$ = mean of yield for sand

Linear Regression Reformulate

Model with dummy variables for soil type:

$$y_j = \beta_0 + \beta_2 x_{2j} + \beta_3 x_{3j} + \epsilon_j, \epsilon_j \sim N(0, \sigma^2)$$

where:

- $x_{2j} = 1$ if soil is loam, else 0
- $x_{3j} = 1$ if soil is sand, else 0 and hence $x_{1j} = 1 - x_{2j} - x_{3j}$
- $\beta_0 = \mu_1$ = intercept (mean of yield for clay)
- $\beta_0 + \beta_2 = \mu_2$ = mean of yield for loam
- $\beta_0 + \beta_3 = \mu_3$ = mean of yield for sand
- same model but with $p = 2$ not $p = 3$.

ANOVA

Let us use two indices instead of dummy variables

- i - indexes the group (soil type): $i \in \{1, 2, 3\}$
- j - indexes observation within group: $j \in \{1, \dots, n_i\}$

Let y_{ij} be the observed yield for observation number $j \in \{1, \dots, n_i\}$ soil type $i \in \{1, \dots, k\}$.

The model (One-way ANOVA):

$$y_{ij} = \mu_i + \epsilon_{ij}$$

where $\epsilon_{ij} \sim N(0, \sigma^2)$ and μ_i is the expected yield for soil type i .

Splitting up total variability

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2, \quad \bar{y}_{..} \text{ is grand mean}$$

For the yield data:

```
## [1] 414.7
```

$$SSA = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{y}_{i.} - \bar{y}_{..})^2 = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2, \quad \bar{y}_{i.} \text{ i-th group mean}$$

For the yield data:

```
## [1] 99.2
```

$$\bullet \text{ SSE} = SST - SSA = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2 = 414.7 - 99.2 = 315.5$$

Between and within group variances:

Dividing Sums of Squares by their respective degrees of freedom gives:

- The between soil type (group) variance:

$$MSA = SSA/(k - 1) = 99.2/2 = 49.6$$

- The within soil type (group) variance:

$$MSE = SSE/(n - k) = 315.5/27 = 11.685$$

where $n = n_1 + n_2 + \cdots + n_k$ is the total number of observations.

Why does it make sense to compare these?

Testing the effect of soil

If there is a soil effect on yield, the expected yields will differ in one way or the other.

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

VS

$$H_1 : \mu_i \neq \mu_{i'}$$

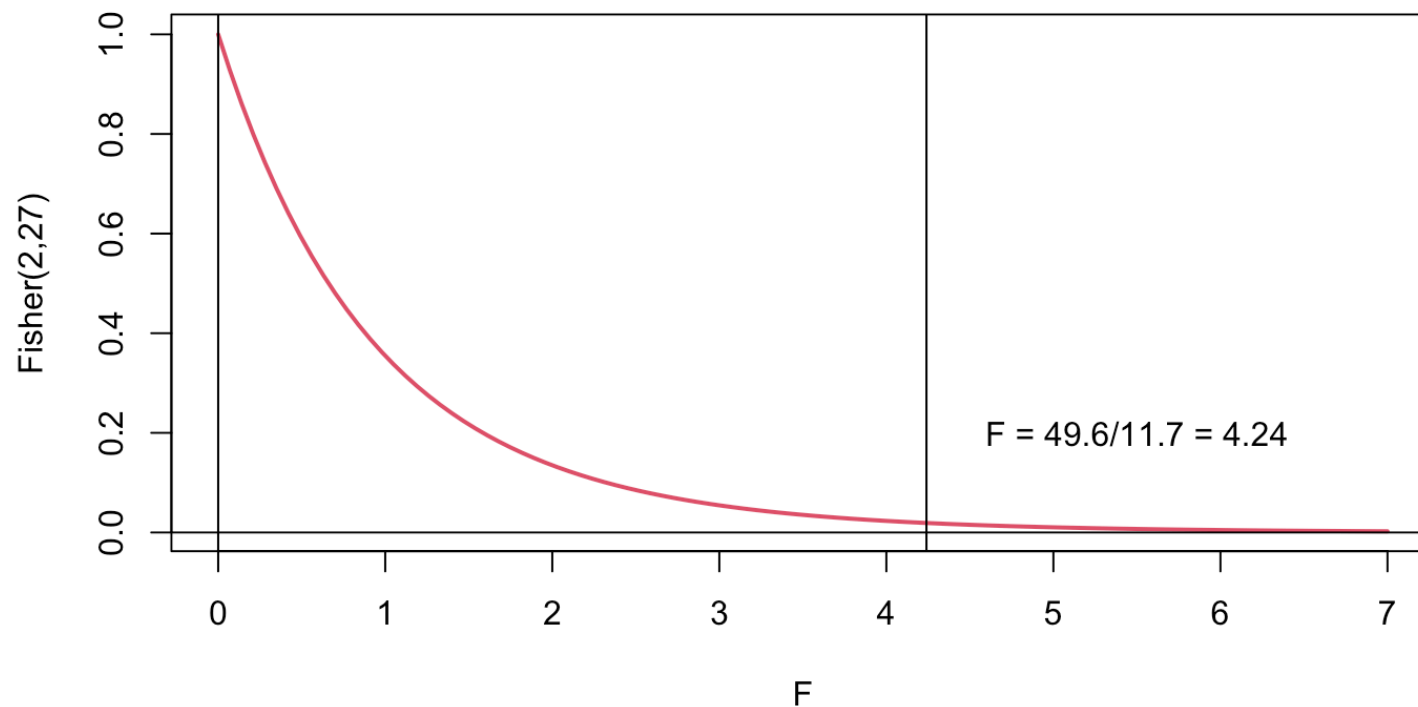
for at least two different soil types i and i' .

Reject H_0 if $MSA \gg MSE$

Test observator, F

If H_0 is true, then

$$F = \frac{MSA}{MSE} \sim F(\nu_1 = k - 1 = 3 - 1, \nu_2 = n - k = 30 - 3)$$



Rejection region

Reject H_0 at test level α if $F_{obs} > F_{\alpha, \nu_1, \nu_2}$, where $\nu_1 = k - 1$ are the degrees of freedom for MSA and $\nu_2 = n - k$ are the degrees of freedom for MSE.

For example, assume a test level of 5%

From a Fisher table we find $F_{0.05, 2, 27} = 3.35$

Alternatively, reject if p-value is smaller than α .

Conclusion?

Estimation of group effects and error variance

The intuitive estimators for μ_i are $\hat{\mu}_i = \bar{y}_i$.

These are also the least squares estimators (for balanced data), which are unbiased and have minimum variance among all unbiased estimators.

As in regression analysis we have $\hat{\sigma}^2 = MSE$.

Splitting up μ_i - i.e. reparametrise

The means may be split up further. Assume

$$\mu_i = \mu + \alpha_i$$

But, all parameters can not be estimated without some extra restriction (choice of parametrization).

- Parametrization 1: Assume $\sum_{i=1}^k \alpha_i = 0$

A “sum to zero” parametrization (“contr.sum” in R)

- Parametrization 2: Assume $\alpha_1 = 0$

A “reference level” or “control level” parametrization (“contr.treatment” in R)

Choice of parametrization in RStudio

Parametrization 1 (Using the lm-function in the mixlm-package)

```
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.sum")
```

Parametrization 2 (Using the lm-function in the mixlm-package)

```
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.treatment")
```

Example: Parametrization 1

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.9      0.6241 19.0673  0.0000
## soil(clay)       -0.4      0.8826 -0.4532  0.6540
## soil(loam)        2.4      0.8826  2.7192  0.0113
```

Compare with group and total means:

```
means <- c(soilmeans, mean(soilmeans))
names(means) <- c("clay", "loam", "sand", "total-mean")
print(means)
```

```
##      clay      loam      sand total-mean
##      11.5      14.3       9.9       11.9
```


Example: Parametrization 2

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	11.5	1.0810	10.6385	0.0000
##	soil(loam)	2.8	1.5287	1.8316	0.0781
##	soil(sand)	-1.6	1.5287	-1.0466	0.3046

Compare with group and total means:

##	clay	loam	sand	total-mean
##	11.5	14.3	9.9	11.9

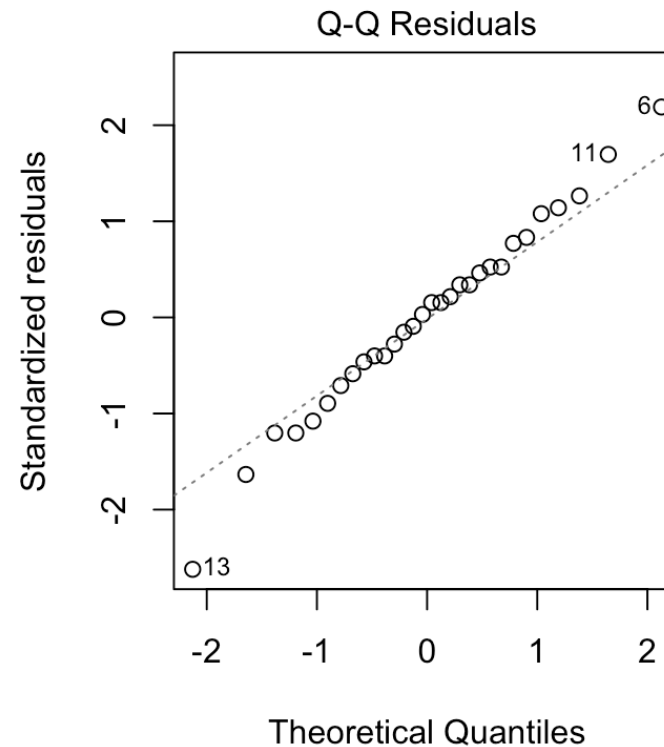
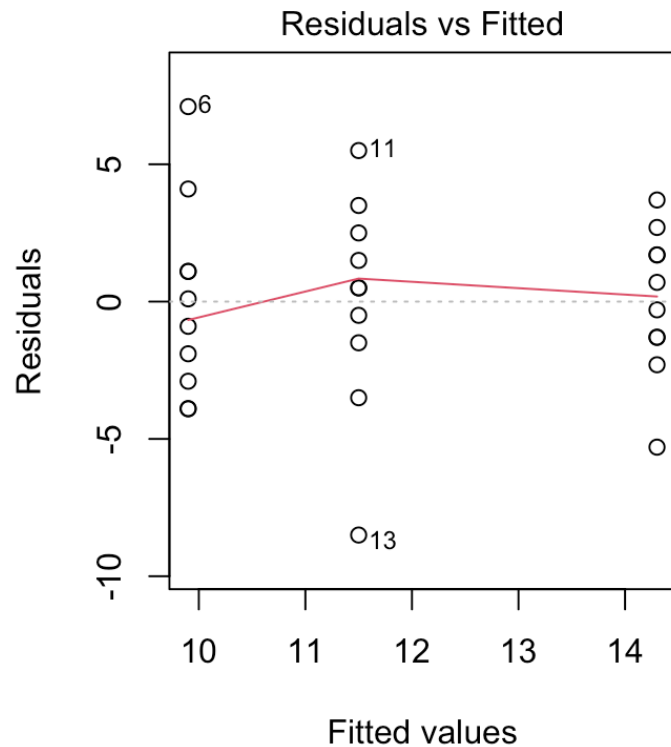
ANOVA table of Sum of squares

This table does not depend on parametrization

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value    Pr(>F)
## soil         2   99.2   49.600    4.2447 0.02495 *
## Residuals  27  315.5   11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual analysis

We should also for ANOVA-models check the assumptions by residual analysis



Pair-wise contrasts

If a factor is significant and has more than two levels: Where are the differences?

May perform a “post hoc” test with contrasts.

A contrasts is a linear function of the factor effects.

$$\theta = c_1\mu_1 + c_2\mu_2 + c_3\mu_3$$

where $\sum_{i=1}^k c_i = 0$

For pair-wise contrasts, only two c_i 's differ from 0.

Example 1 (With parametrization 1)

Comparing soil type “loam” (level 1) with “clay” (level 2):

$$\theta_1 = 1\mu_1 + (-1)\mu_2 + 0\mu_3$$

$$= 1(\mu + \alpha_1) - 1(\mu + \alpha_2) = \alpha_1 - \alpha_2$$

$$\text{Estimate: } \hat{\theta}_1 = \hat{\alpha}_1 - \hat{\alpha}_2 = -0.4 - 2.4 = -2.8$$

```
gmodels::fit.contrast(model=mod1, varname='soil', df=TRUE, coeff=c(1, -1, 0), conf.int
```

```
##              Estimate Std. Error  t value  Pr(>|t|) DF  lower CI
## soil c=( 1 -1 0 )    -2.8    1.528737 -1.831577 0.07806971 27 -5.936709
##              upper CI
## soil c=( 1 -1 0 ) 0.3367094
```

Example 2: “loam” vs “sand”

Define required pairwise contrast:

$$\theta_2 = 0\mu_1 + 1\mu_2 - 1\mu_3 = 1(\mu + \alpha_2) - 1(\mu + \alpha_3) = \alpha_2 - \alpha_3$$

Estimation

$$\hat{\alpha}_3 = -(-0.4 + 2.4) = -2 \quad \hat{\theta}_2 = \hat{\alpha}_2 - \hat{\alpha}_3 = 2.4 - (-2) = 4.4$$

##		Estimate	Std. Error	t value	Pr(> t)	DF	lower CI	upper CI
##	soil c=(0 1 -1)	4.4	1.528737	2.878193	0.007728022	27	1.263291	7.536709

Multiple testing - Tukey test

Type 1: $\alpha = P(\text{Rejecting } H_0 \mid H_0 \text{ is TRUE})$

If there are many levels of a factor, a Tukey test is preferable.

Controls overall error rate (family-wise error) of all pairwise tests.

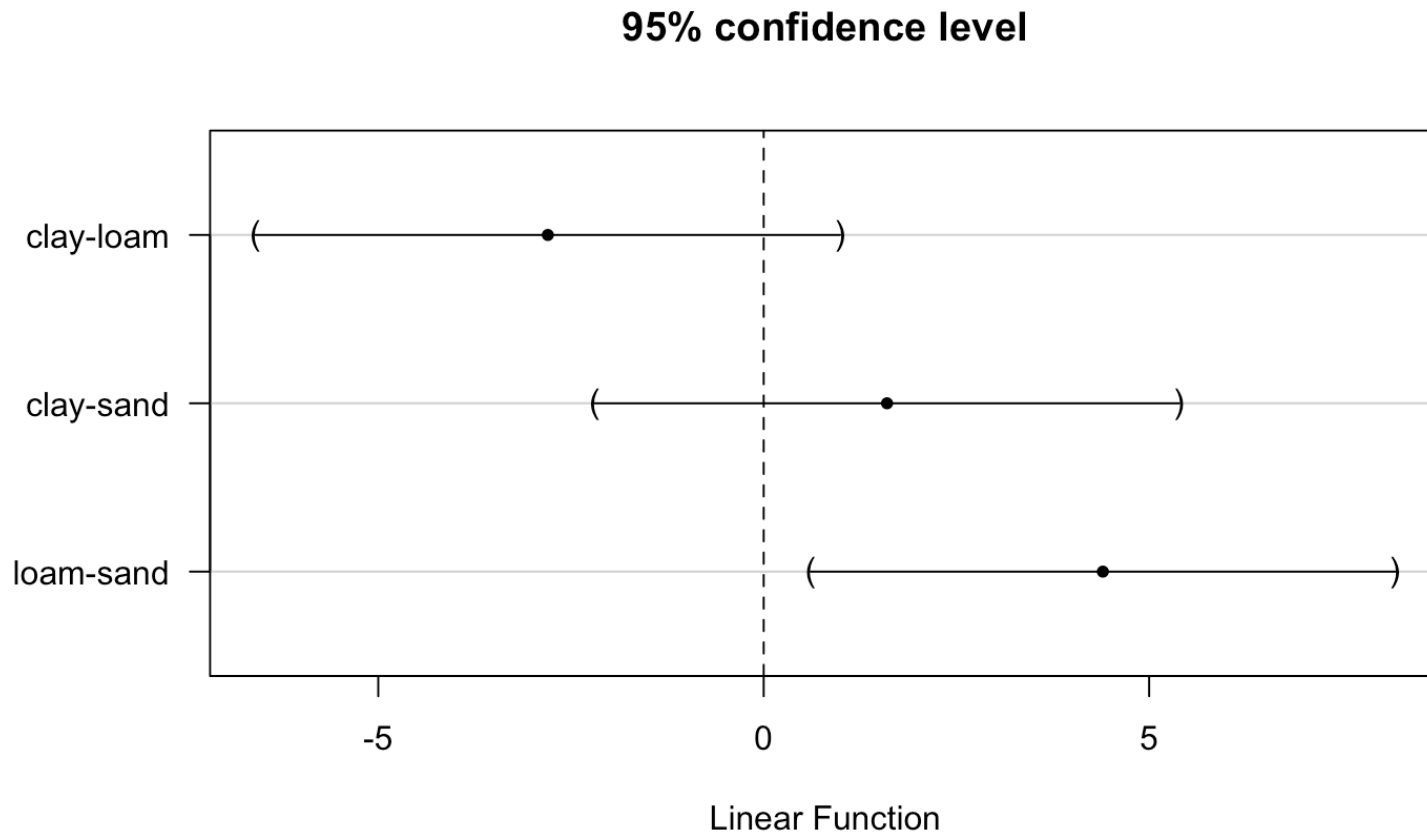
Use 'simple.glht'-function in the mixlm-package.

```
ptests <- mixlm::simple.glht(mod1, "soil", corr = c("Tukey"), level = 0.95)
print(ptests)
plot(ptests)
```

R Commander output - Tukey

```
##
## Simultaneous Confidence Intervals and Tests for General Linear Hypotheses
##
## Multiple Comparisons of Means: Tukey Contrasts
##
##
## Fit: stats::lm(formula = yield ~ soil, data = yields)
##
## Quantile = 2.4794
## Minimum significant difference = 3.7904
## 95% confidence level
##
## Linear Hypotheses:
##           Lower Center   Upper Std.Err t value  P(>t)
## clay-loam -6.5904 -2.8000  0.9904  1.5287  -1.832 0.1785
## clay-sand  -2.1904  1.6000  5.3904  1.5287   1.047 0.5546
## loam-sand   0.6096  4.4000  8.1904  1.5287   2.878 0.0204 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
```


Pairwise confidence intervals plot



The ANOVA applet

The ANOVA “idea”: Studying means by splitting variability

[ANOVA shiny app](#)

Topics for next lecture

- ANOVA with several factors
- The factorial applet
- Effect plots
- Estimation and hypothesis testing
- Tukey test and confidence intervals
- ANCOVA

Announcements

- Elias is travelling, online session this Thursday

-What do we do with Mondays

- Assignment 1 will be published as a quiz next Tuesday, deadline March 10. 70% needed to pass.
- Practice quiz on canvas!